

Quick sign test requested by Luca

Ratio test for Eqs. (29) and (32), using real values of z

Short verdict. The stochastic shortcut matrix, where $\lambda = \beta(1 - q)$ is moved from the self-loop at u to the directed edge $u \rightarrow v$, agrees with Eq. (32) only with the corrected plus/plus sign. The original minus/minus sign does not match this stochastic matrix.

I used Luca's suggested ratio check: compute the first-passage generating function from a full-ring shortcut propagator by

$$\tilde{F}_{n_0}(v, z) = \frac{\tilde{S}_{n_0}(v, z)}{\tilde{S}_v(v, z)}.$$

I then compared this value with Eq. (32) using both signs.

There is one important point in the test. If Eq. (29) is used literally as printed in the manuscript, its ratio agrees with the original minus/minus Eq. (32). However, the actual probability-conserving stochastic shortcut matrix for $u \rightarrow v$ agrees with the corrected plus/plus Eq. (32). Thus this test pushes the sign inconsistency one step earlier: Eq. (29), as printed, has the opposite defect sign if it is interpreted as moving self-loop mass from u to v .

Sanity check. For positive real $z < 1$, a first-passage generating function has non-negative coefficients and therefore cannot be negative. In the $N = 10$, $q = 2/3$, $\beta = 2/7$, $u = 5$, $v = 0$ test, the original sign gives a negative value for $n_0 = 4$, $z = 0.8$, while the stochastic matrix and the corrected sign give a positive value.

Error Summary

The grid uses two symmetric cases, including Luca's figure parameters $N = 10$, $q = 2/3$, $\beta = 2/7$, $u = 5$, $v = 0$, with several starts n_0 and real values of z . Site labels here are zero-based.

Quantity compared over the grid				Maximum absolute error
Original Eq. (32) vs stochastic shortcut matrix				3.669746×10^{-1}
Corrected Eq. (32) vs stochastic shortcut matrix				2.775558×10^{-16}
Literal Eq. (29) ratio vs original Eq. (32)				1.318390×10^{-16}
Stochastic-sign Eq. (29) ratio vs corrected Eq. (32)				5.551115×10^{-17}
Eq. (19) $W_{n_0}(u, z)$ vs absorbing matrix				1.043610×10^{-14}
Eq. (28) $W_u(u, z)$ vs absorbing matrix				1.687539×10^{-14}

Representative Values

n_0	z	stochastic matrix	literal Eq. (29)	original Eq. (32)	corrected Eq. (32)
2	0.50	0.044096501039	0.042961443311	0.042961443311	0.044096501039
2	0.80	0.197025316799	0.173752283343	0.173752283343	0.197025316799
2	0.95	0.532769134231	0.394627514541	0.394627514541	0.532769134231
4	0.80	0.095953324694	-0.033502923905	-0.033502923905	0.095953324694

Reproducibility

The script used to generate the numbers is

```
scripts/luca_quick_sign_test.py.
```

It writes the full numerical table to

```
luca_quick_sign_test_results.csv.
```